

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05

Total Marks:100

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Shaik Mohammed Abrar / 192472185 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

RUBRICS FOR CALCULATION

TECHNICAL PRESENTATION					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Technical Presentation

Choose ANY ONE Scenario:

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage. Explain how:

- A. Python programming basics are used
- B. Pandas and NumPy are used for data handling
- C. Matplotlib and Seaborn are used for result visualization

2. Disease Prediction System

A hospital is planning to build a simple disease prediction system using patient details such as age, blood pressure, sugar level, and cholesterol. Explain how:

- A. Python libraries support medical data analysis
- B. Scikit-Learn is used for model training and prediction
- C. Jupyter Notebook or Google Colab helps in implementation

3. Sales Forecasting Dashboard

A retail company wants to analyze monthly sales data and predict future sales trends. Explain how:

- A. Pandas is used for reading and cleaning sales data
- B. SciPy and Scikit-Learn are used for analysis and prediction
- C. Matplotlib and Seaborn are used to create business visualizations

4. Image Classification Application

A startup is developing an image classification application to identify food items from images. Explain how:

- A. Python is used as the development language
- B. TensorFlow and PyTorch are used for deep learning model development
- C. Notebook environments help in training, testing, and visualizing results

5. Smart Agriculture Data Monitoring

An agriculture research team collects temperature, humidity, soil moisture, and crop growth data from sensors. Explain how:

- A. NumPy is used for numerical computation
- B. Pandas is used for organizing sensor data
- C. Scikit-Learn and visualization libraries support crop condition analysis

TECHNICAL PRESENTATION

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage. Explain how:

- A. Python programming basics are used
- B. Pandas and NumPy are used for data handling
- C. Matplotlib and Seaborn are used for result visualization

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling.

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage.

A. Python Programming Basics

Python provides the foundation for developing the Student Performance Analysis System.

1. **Variables** store student details such as name, marks, attendance, and pass percentage.
2. **Data types** like int, float, and string are used to represent student information.
3. **Conditional statements (if-else)** determine whether a student has passed or failed based on marks.
4. **Loops (for, while)** process records of multiple students.
5. **Functions** improve code reusability by calculating average marks, attendance percentage, and pass percentage.
6. **File handling** is used to read and save student data in CSV or text files.

B. Pandas and NumPy for Data Handling

Pandas

Pandas is used to organize and preprocess student data.

1. Load student records from CSV using `pd.read_csv()`.
2. Handle missing values using `fillna()` or `dropna()`.
3. Remove duplicate records using `drop_duplicates()`.

4. Filter students based on attendance or marks.
5. Calculate average marks, highest marks, lowest marks, and pass percentage.
6. Store data in a **DataFrame** for easy analysis.

NumPy

NumPy performs fast numerical computations.

1. Store marks as arrays.
2. Calculate statistical measures such as:
 - Mean
 - Median
 - Standard Deviation
 - Variance
3. Perform mathematical operations efficiently on large datasets.

C. Matplotlib and Seaborn for Result Visualization

Matplotlib

Matplotlib is used to create basic graphs for student performance.

1. **Bar Chart:** Compare marks of students.
2. **Line Graph:** Show attendance trends over time.
3. **Pie Chart:** Display pass and fail percentages.
4. **Histogram:** Show distribution of student marks.

Seaborn

Seaborn provides advanced statistical visualizations.

1. **Count Plot:** Number of students who passed and failed.
2. **Box Plot:** Detect outliers in student marks.
3. **Heatmap:** Show correlation between marks and attendance.
4. **Scatter Plot:** Analyze the relationship between attendance and marks.

Workflow of the System

1. Collect student marks and attendance.
2. Load data using Pandas.
3. Clean data by removing missing and duplicate values.
4. Use NumPy to calculate statistical measures.
5. Calculate pass percentage and average marks.
6. Visualize results using Matplotlib and Seaborn.
7. Generate reports to help teachers analyze student performance.

Python Libraries Used

Library	Purpose
Pandas	Data loading, cleaning, and analysis
NumPy	Numerical and statistical calculations
Matplotlib	Basic charts and graphs
Seaborn	Advanced statistical visualizations

Expected Outcome

The Student Performance Analysis System efficiently preprocesses student data, calculates performance statistics, and generates meaningful visualizations. This helps faculty identify high-performing students, students with low attendance, and overall class performance for better academic decision-making.

Code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns
```

```
df = pd.read_csv("Student-AT2.csv")

print("\nSTUDENT PERFORMANCE DATA\n")

print(df)

print("\nFIRST 5 RECORDS\n")

print(df.head())

print("\nLAST 5 RECORDS\n")

print(df.tail())

print("\nDATA INFORMATION\n")

print(df.info())

print("\nMISSING VALUES\n")

print(df.isnull().sum())

df = df.drop_duplicates()

df["Attendance"] = df["Attendance"].fillna(df["Attendance"].mean())

df["Assignment_Marks"] = df["Assignment_Marks"].fillna(df["Assignment_Marks"].mean())

df["Internal_Marks"] = df["Internal_Marks"].fillna(df["Internal_Marks"].mean())

df["Marks"] = df["Marks"].fillna(df["Marks"].mean())

df["Result"] = np.where(df["Marks"] >= 40, "Pass", "Fail")

marks = df["Marks"].to_numpy()

attendance = df["Attendance"].to_numpy()

print("\nSTATISTICAL ANALYSIS\n")

print("Mean Marks :", np.mean(marks))

print("Median Marks :", np.median(marks))

print("Standard Deviation :", np.std(marks))

print("Variance :", np.var(marks))

print("Maximum Marks :", np.max(marks))
```

```
print("Minimum Marks :", np.min(marks))

print("\nAttendance Statistics\n")

print("Mean Attendance :", np.mean(attendance))

print("Maximum Attendance :", np.max(attendance))

print("Minimum Attendance :", np.min(attendance))

print("\nMode of Marks")

print(df["Marks"].mode())

print("\nDepartment Wise Average Marks")

print(df.groupby("Department")["Marks"].mean())

print("\nSemester Wise Average Marks")

print(df.groupby("Semester")["Marks"].mean())

total_students = len(df)

passed_students = len(df[df["Result"] == "Pass"])

failed_students = len(df[df["Result"] == "Fail"])

pass_percentage = (passed_students / total_students) * 100

fail_percentage = (failed_students / total_students) * 100

print("\nTotal Students :", total_students)

print("Passed Students :", passed_students)

print("Failed Students :", failed_students)

print("Pass Percentage : {:.2f}%".format(pass_percentage))

print("Fail Percentage : {:.2f}%".format(fail_percentage))

print("\nTop Performer")

print(df.loc[df["Marks"].idxmax()])

print("\nStudents with Attendance Below 75%")

print(df[df["Attendance"] < 75])
```

```
sns.set_style("whitegrid")

plt.figure(figsize=(12,6))

plt.bar(df["Name"], df["Marks"])

plt.title("Student Marks")

plt.xlabel("Student Name")

plt.ylabel("Marks")

plt.xticks(rotation=90)

plt.show()

plt.figure(figsize=(12,6))

plt.plot(df["Name"], df["Attendance"], marker='o')

plt.title("Attendance of Students")

plt.xlabel("Student Name")

plt.ylabel("Attendance")

plt.xticks(rotation=90)

plt.show()

result = df["Result"].value_counts()

plt.figure(figsize=(6,6))

plt.pie(result.values, labels=result.index, autopct="%1.1f%%", startangle=90)

plt.title("Pass and Fail Percentage")

plt.show()

plt.figure(figsize=(8,5))

plt.hist(df["Marks"], bins=10)

plt.title("Marks Distribution")

plt.xlabel("Marks")

plt.ylabel("Frequency")
```



```
plt.show()

plt.figure(figsize=(8,5))

plt.scatter(df["Attendance"], df["Marks"])

plt.title("Attendance vs Marks")

plt.xlabel("Attendance")

plt.ylabel("Marks")

plt.show()

plt.figure(figsize=(6,5))

sns.boxplot(y=df["Marks"])

plt.title("Box Plot of Marks")

plt.show()

plt.figure(figsize=(6,5))

sns.countplot(x="Department", data=df)

plt.title("Department Wise Student Count")

plt.show()

plt.figure(figsize=(6,5))

sns.countplot(x="Result", data=df)

plt.title("Pass and Fail Count")

plt.show()

plt.figure(figsize=(6,5))

sns.barplot(x="Department", y="Marks", data=df)

plt.title("Average Marks by Department")

plt.show()

plt.figure(figsize=(6,5))

sns.boxplot(x="Department", y="Marks", data=df)
```

```

plt.title("Department Wise Marks Distribution")

plt.show()

plt.figure(figsize=(6,5))

sns.scatterplot(x="Attendance", y="Marks", hue="Department", data=df)

plt.title("Attendance vs Marks by Department")

plt.show()

plt.figure(figsize=(6,5))

sns.heatmap(df[["Attendance", "Assignment_Marks", "Internal_Marks", "Marks"]].corr(),
annot=True, cmap="coolwarm")

plt.title("Correlation Heatmap")

plt.show()

print("\nAnalysis Completed Successfully.")

```

Output 1:

STUDENT PERFORMANCE DATA

	Student_ID	Name	Department	Semester	Attendance	Assignment_Marks
0	1001	Aarav	CSE	3	68	10
1	1002	Aditi	AI	5	71	17
2	1003	Rohan	IT	7	74	13
3	1004	Priya	ECE	3	77	20
4	1005	Rahul	CSE	5	80	16
..	...	...	...	...	...	...
95	1096	Varsha	ECE	7	74	15
96	1097	Yogesh	CSE	3	77	11
97	1098	Zainab	AI	5	80	18
98	1099	Aravind	IT	7	83	14
99	1100	Bhargavi	ECE	3	86	10

	Internal_Marks	Marks	Result
0	14	59	Pass
1	19	72	Pass
2	24	74	Pass
3	29	87	Pass
4	17	72	Pass
..	...	...	...
95	30	87	Pass
96	18	64	Pass
97	23	77	Pass
98	28	79	Pass
99	16	64	Pass

[100 rows x 9 columns]

FIRST 5 RECORDS

	Student_ID	Name	Department	Semester	Attendance	Assignment_Marks	\
0	1001	Aarav	CSE	3	68	10	
1	1002	Aditi	AI	5	71	17	
2	1003	Rohan	IT	7	74	13	
3	1004	Priya	ECE	3	77	20	
4	1005	Rahul	CSE	5	80	16	

	Internal_Marks	Marks	Result
0	14	59	Pass
1	19	72	Pass
2	24	74	Pass
3	29	87	Pass
4	17	72	Pass

LAST 5 RECORDS

	Student_ID	Name	Department	Semester	Attendance	Assignment_Marks	\
95	1096	Varsha	ECE	7	74	15	
96	1097	Yogesh	CSE	3	77	11	
97	1098	Zainab	AI	5	80	18	
98	1099	Aravind	IT	7	83	14	
99	1100	Bhargavi	ECE	3	86	10	

	Internal_Marks	Marks	Result
95	30	87	Pass
96	18	64	Pass
97	23	77	Pass
98	28	79	Pass
99	16	64	Pass

DATA INFORMATION

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Student_ID            100 non-null   int64
1   Name                  100 non-null   object
2   Department            100 non-null   object
3   Semester              100 non-null   int64
4   Attendance            100 non-null   int64
5   Assignment_Marks      100 non-null   int64
6   Internal_Marks        100 non-null   int64
7   Marks                 100 non-null   int64
8   Result                100 non-null   object
dtypes: int64(6), object(3)
memory usage: 7.2+ KB
None

```

MISSING VALUES

Student_ID	0
Name	0
Department	0
Semester	0
Attendance	0

Assignment_Marks 0
Internal_Marks 0
Marks 0
Result 0
dtype: int64

STATISTICAL ANALYSIS

Mean Marks : 75.34
Median Marks : 75.0
Standard Deviation : 6.62
Variance : 43.8244
Maximum Marks : 90
Minimum Marks : 59

Attendance Statistics

Mean Attendance : 82.58
Maximum Attendance : 98
Minimum Attendance : 68

Mode of Marks

0 72
Name: Marks, dtype: int64

Department Wise Average Marks

Department
AI 74.68
CSE 73.04
ECE 77.08
IT 76.56
Name: Marks, dtype: float64

Semester Wise Average Marks

Semester
3 75.235294
5 75.454545
7 75.333333
Name: Marks, dtype: float64

Total Students : 100
Passed Students : 100
Failed Students : 0
Pass Percentage : 100.00%
Fail Percentage : 0.00%

Top Performer

Student_ID 1048
Name Anjali
Department ECE
Semester 7
Attendance 85
Assignment_Marks 20
Internal_Marks 28
Marks 90
Result Pass
Name: 47, dtype: object

Students with Attendance Below 75%

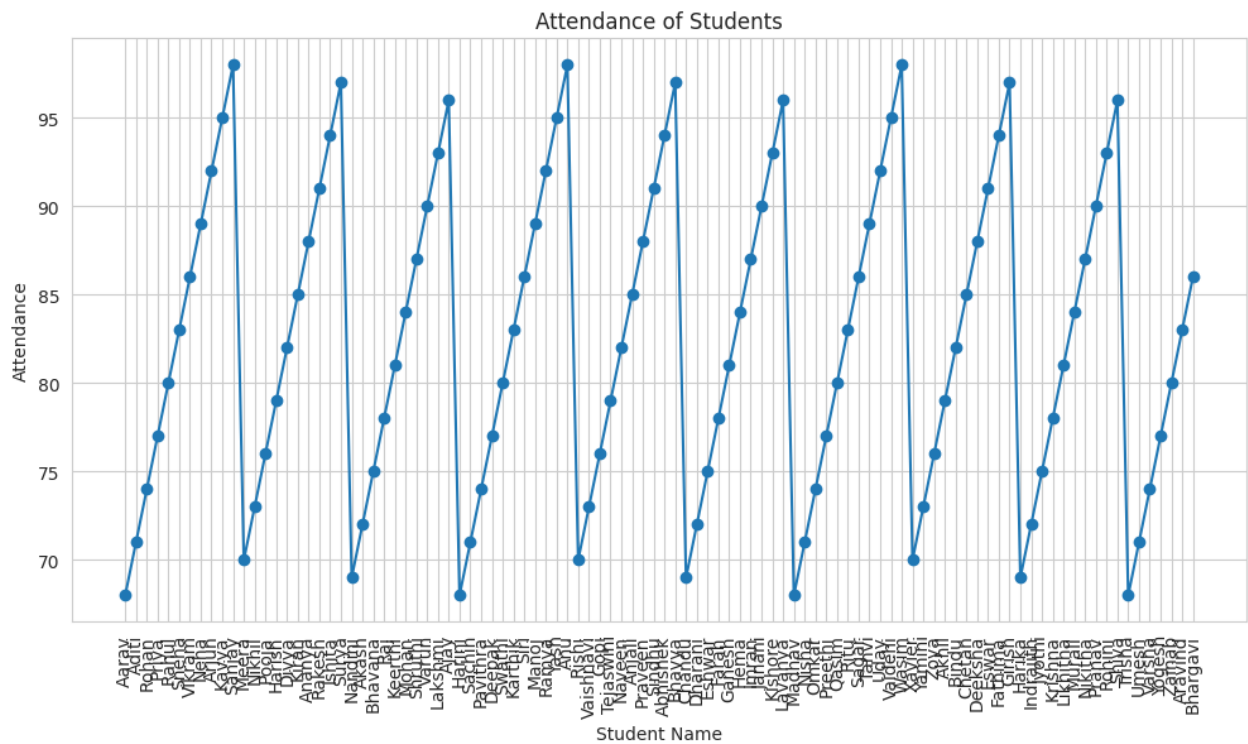
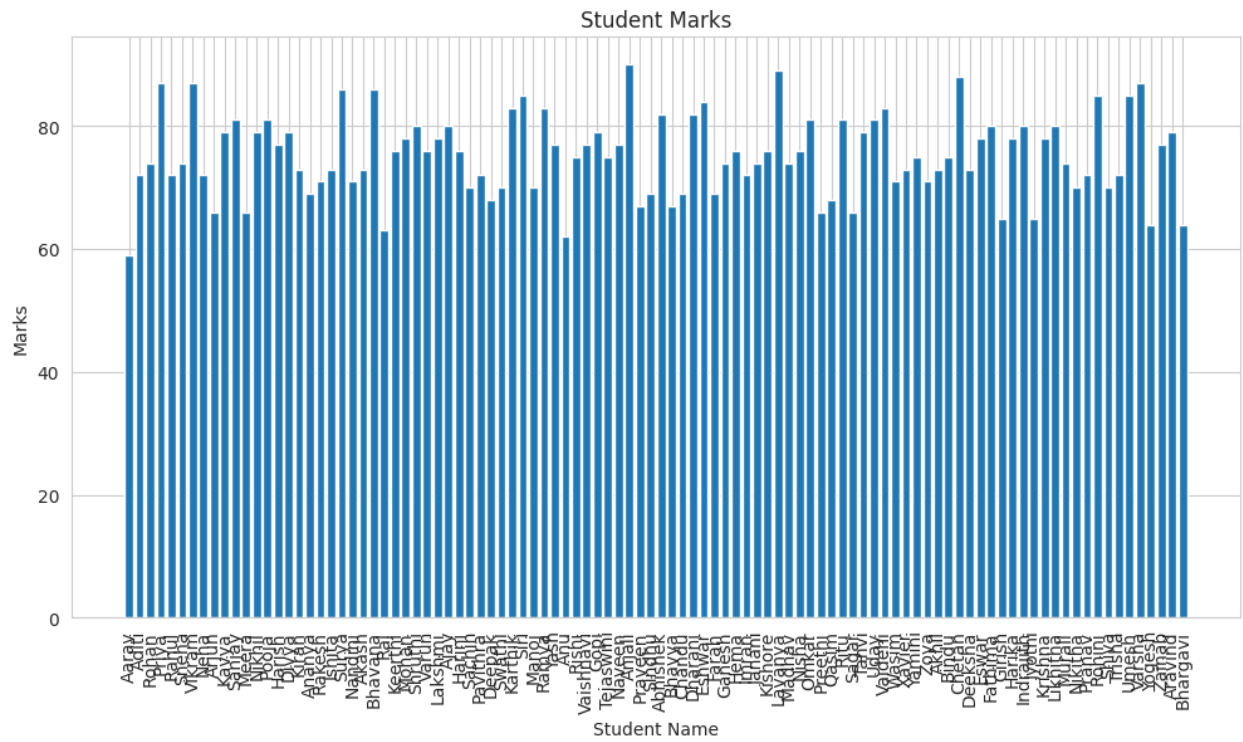
Student_ID	Name	Department	Semester	Attendance	Assignment_Marks
\					

0	1001	Aarav	CSE	3	68
10					
1	1002	Aditi	AI	5	71
17					
2	1003	Rohan	IT	7	74
13					
11	1012	Meera	ECE	7	70
10					
12	1013	Nikhil	CSE	3	73
17					
21	1022	Nandini	AI	3	69
14					
22	1023	Akash	IT	5	72
10					
31	1032	Harini	ECE	5	68
18					
32	1033	Sachin	CSE	7	71
14					
33	1034	Pavithra	AI	3	74
10					
42	1043	Rishi	IT	3	70
18					
43	1044	Vaishnavi	ECE	5	73
14					
52	1053	Chandu	CSE	5	69
11					
53	1054	Dharani	AI	7	72
18					
62	1063	Madhav	IT	7	68
15					
63	1064	Nisha	ECE	3	71
11					
64	1065	Omkar	CSE	5	74
18					
73	1074	Xavier	AI	5	70
15					
74	1075	Yamini	IT	7	73
11					
83	1084	Harika	ECE	7	69
19					
84	1085	Indrajith	CSE	3	72
15					
93	1094	Trisha	AI	3	68
12					
94	1095	Umesh	IT	5	71
19					
95	1096	Varsha	ECE	7	74
15					

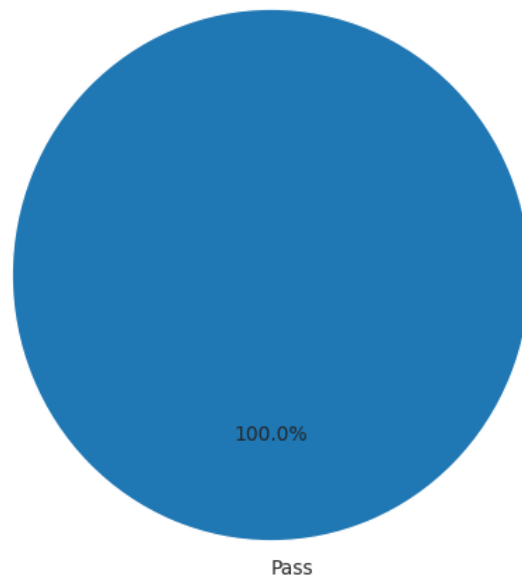
	Internal_Marks	Marks	Result
0	14	59	Pass
1	19	72	Pass
2	24	74	Pass
11	18	66	Pass
12	23	79	Pass
21	17	71	Pass
22	22	73	Pass
31	16	76	Pass
32	21	70	Pass
33	26	72	Pass
42	20	75	Pass

43	25	77	Pass
52	19	69	Pass
53	24	82	Pass
62	18	74	Pass
63	23	76	Pass
64	28	81	Pass
73	22	73	Pass
74	27	75	Pass
83	21	78	Pass
84	26	80	Pass
93	20	72	Pass
94	25	85	Pass
95	30	87	Pass

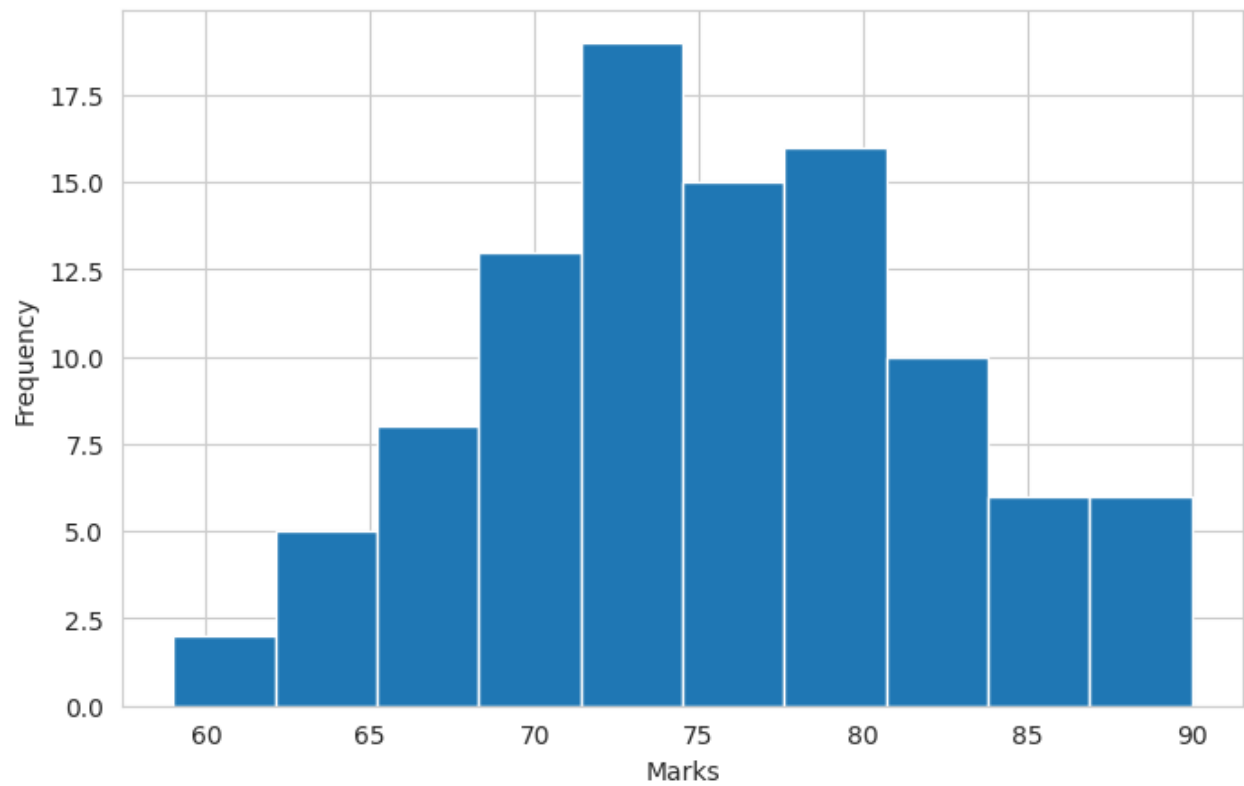
Analysis Completed Successfully.

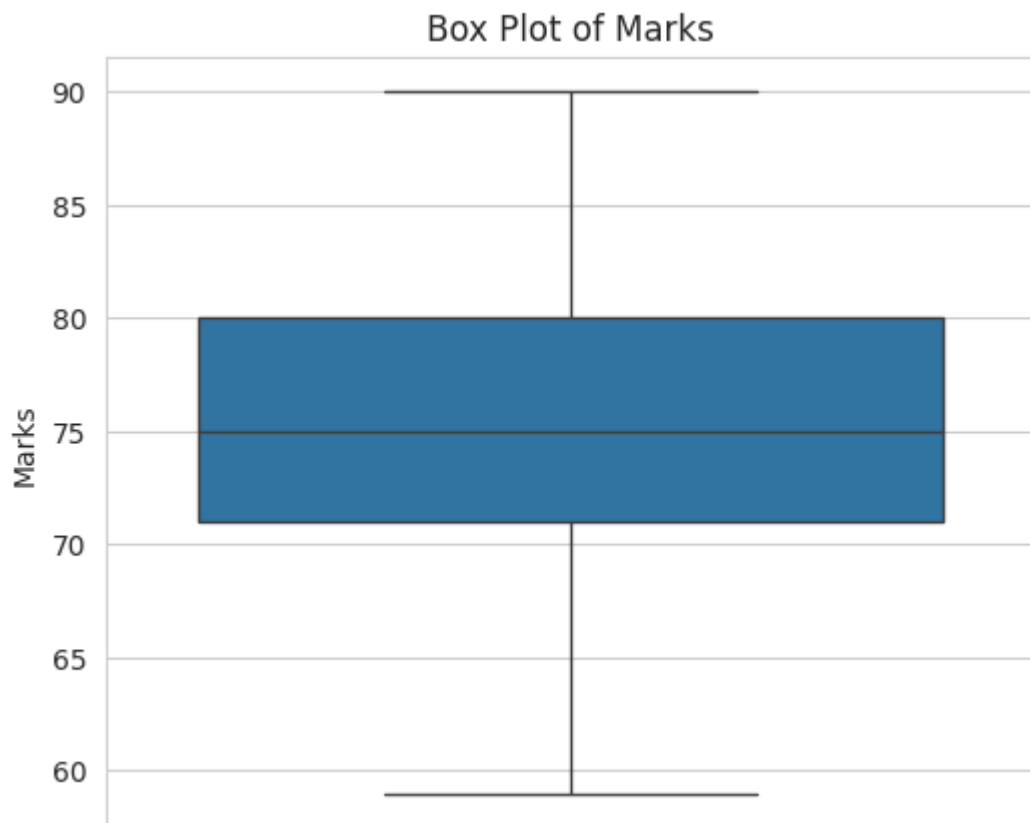
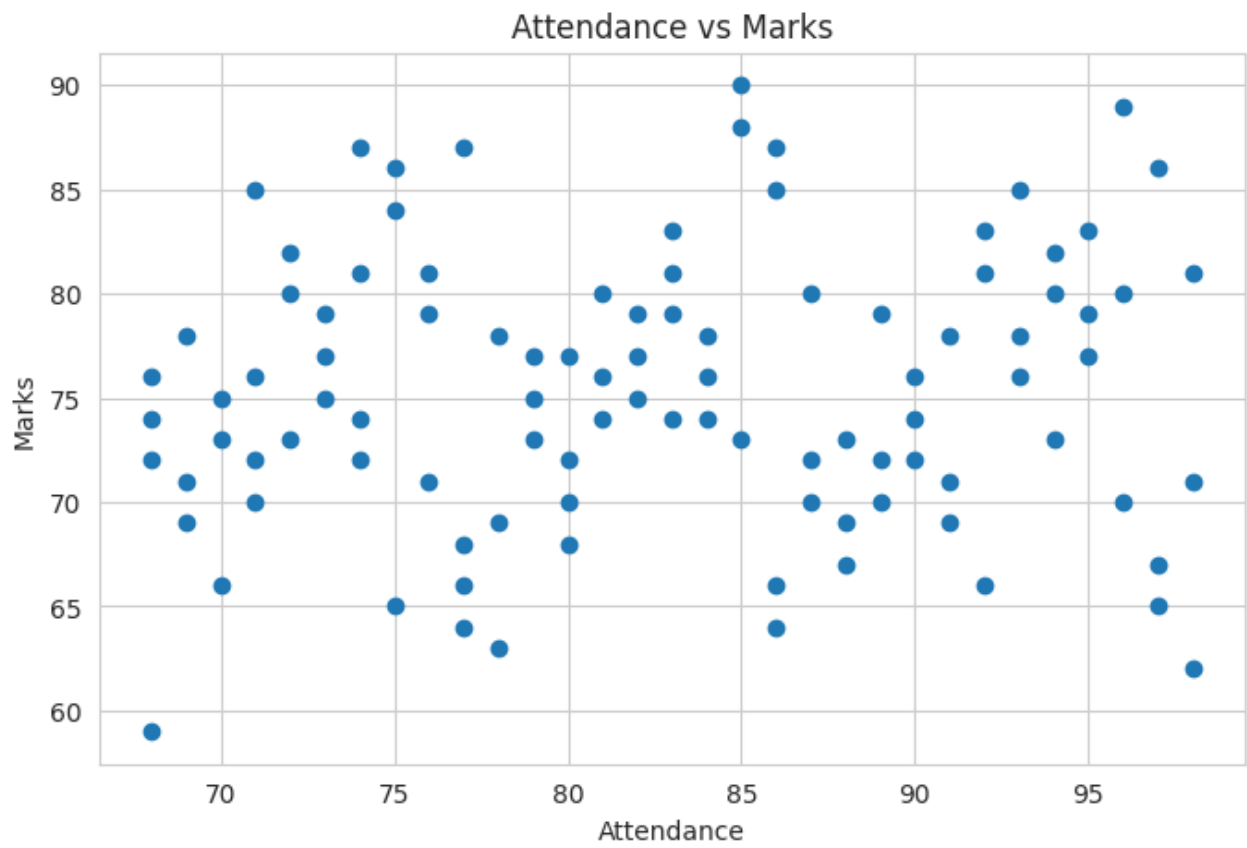


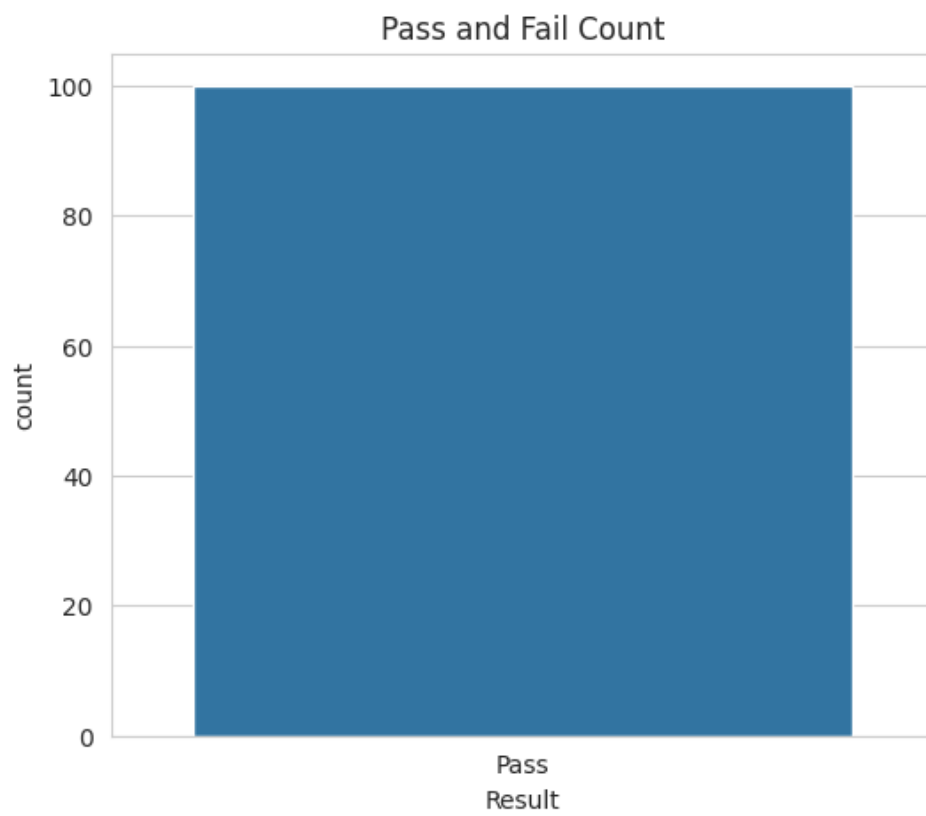
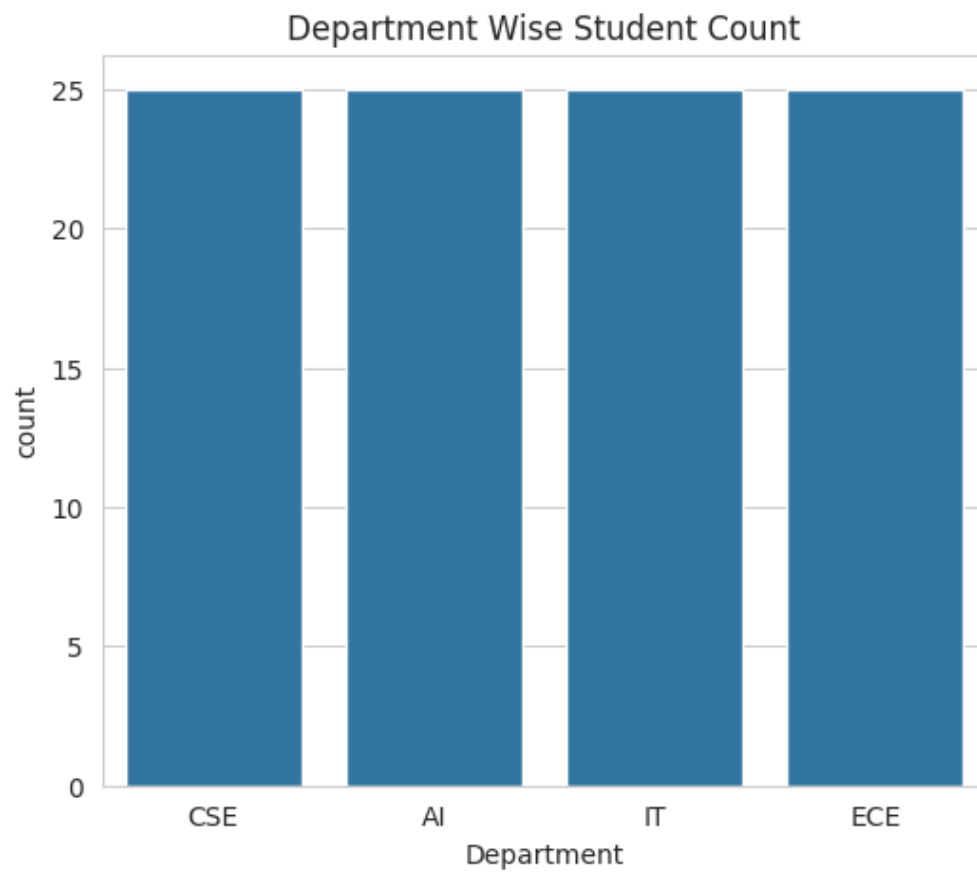
Pass and Fail Percentage

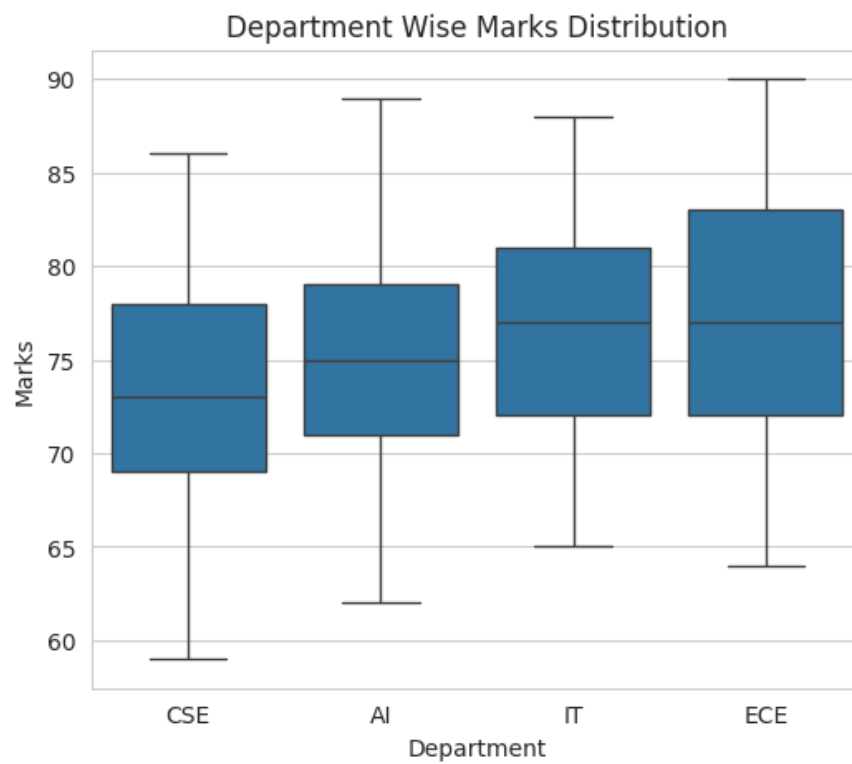
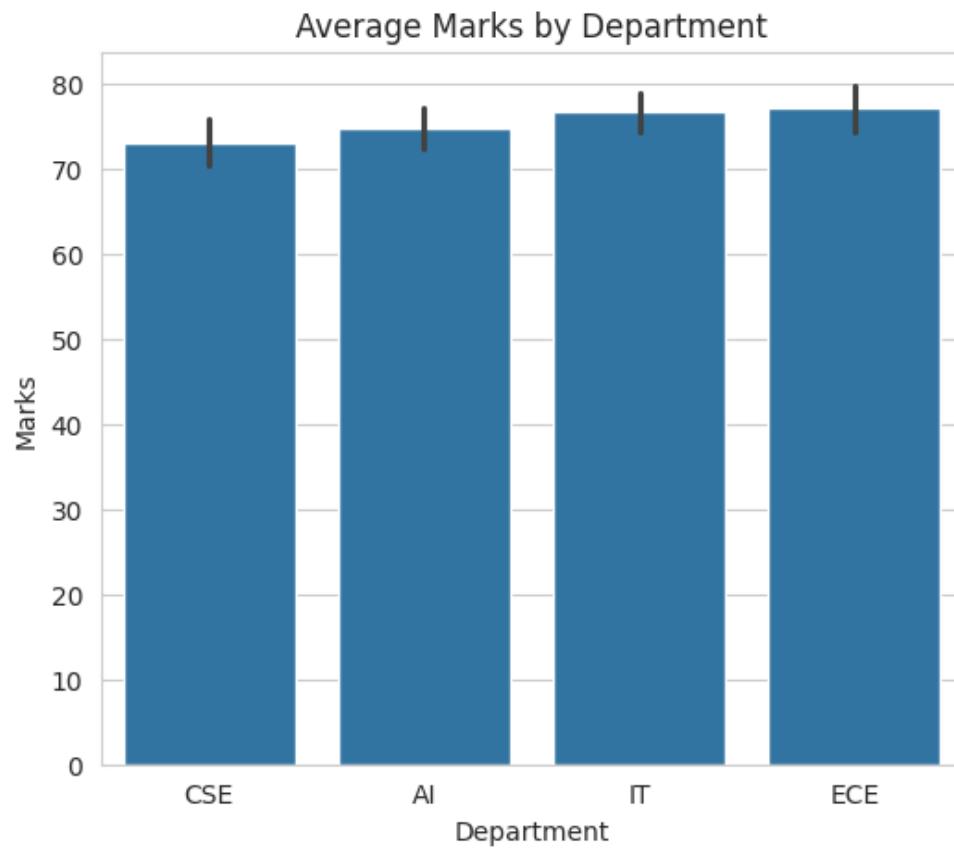


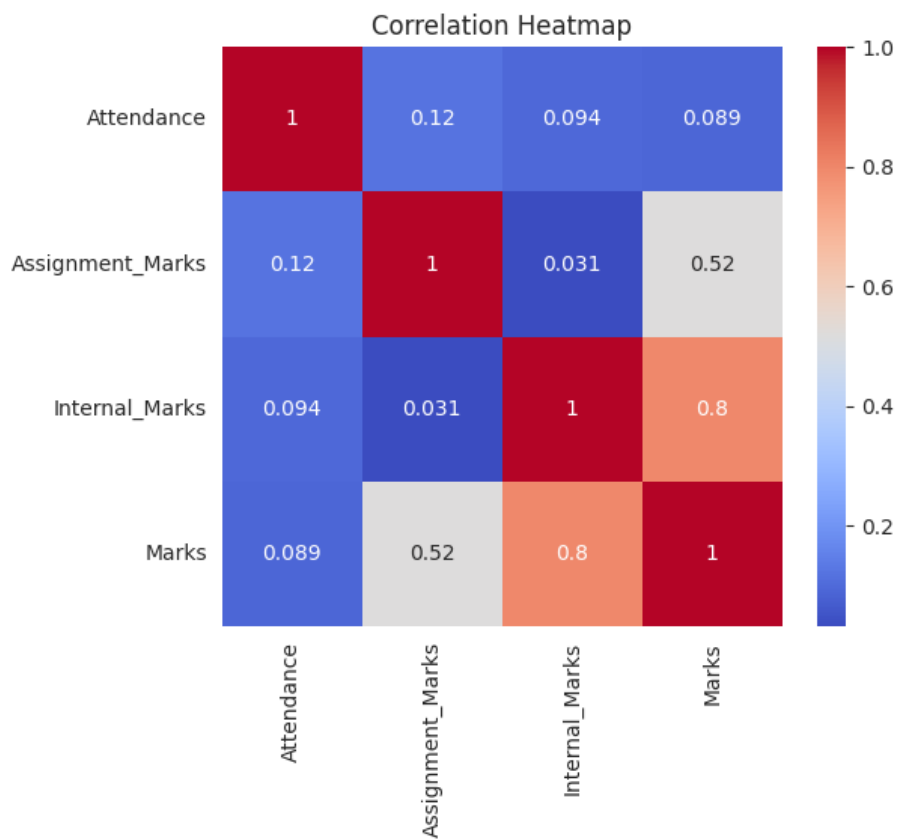
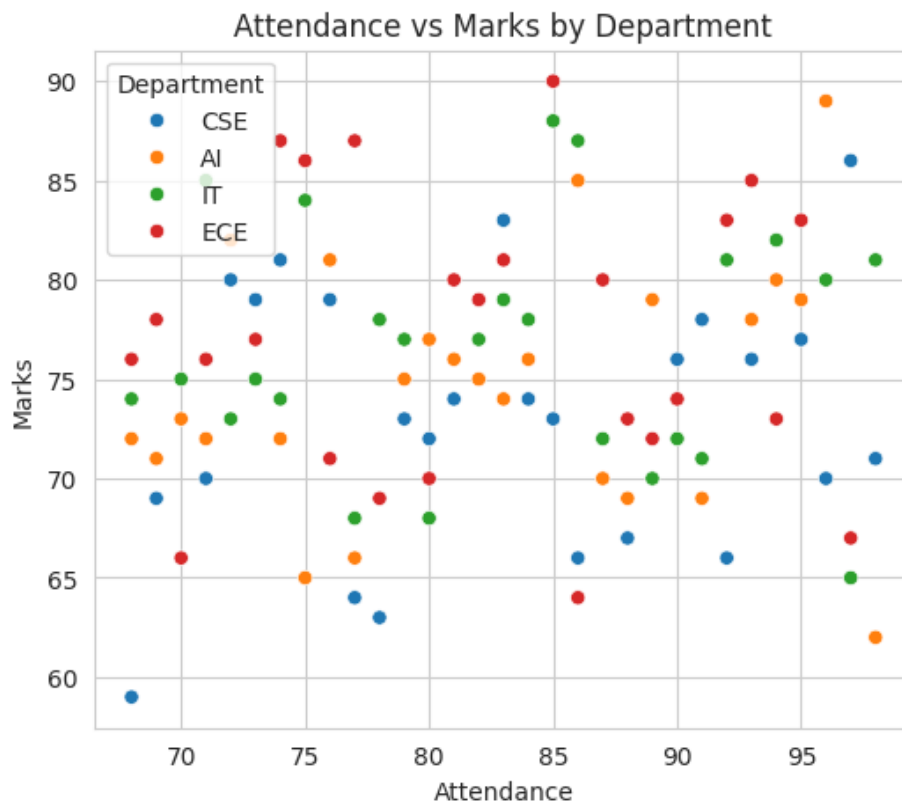
Marks Distribution











Output 2:

STUDENT PERFORMANCE DATA

```

Student_ID      Name Department  Semester  Attendance  Assignment_Marks
\

```

0	2001	Aarav	CSE	3	95	9
1	2002	Aditi	AI	5	70	11
2	2003	Rohan	IT	7	98	19
3	2004	Priya	ECE	3	92	14
4	2005	Rahul	CSE	5	60	11
..	...	...	...	...	...	...
95	2096	Varsha	ECE	7	84	17
96	2097	Yogesh	CSE	3	59	16
97	2098	Zainab	AI	5	71	10
98	2099	Aravind	IT	7	59	11
99	2100	Bhargavi	ECE	3	65	15

	Internal_Marks	Marks	Result
0	10	26	Fail
1	14	21	Fail
2	27	20	Fail
3	11	18	Fail
4	17	34	Fail
..	...	...	...
95	28	46	Pass
96	16	72	Pass
97	21	96	Pass
98	21	58	Pass
99	27	85	Pass

[100 rows x 9 columns]

FIRST 5 RECORDS

	Student_ID	Name	Department	Semester	Attendance	Assignment_Marks	\
0	2001	Aarav	CSE	3	95	9	
1	2002	Aditi	AI	5	70	11	
2	2003	Rohan	IT	7	98	19	
3	2004	Priya	ECE	3	92	14	
4	2005	Rahul	CSE	5	60	11	

	Internal_Marks	Marks	Result
0	10	26	Fail
1	14	21	Fail
2	27	20	Fail
3	11	18	Fail
4	17	34	Fail

LAST 5 RECORDS

	Student_ID	Name	Department	Semester	Attendance	Assignment_Marks	\
95	2096	Varsha	ECE	7	84	17	
96	2097	Yogesh	CSE	3	59	16	
97	2098	Zainab	AI	5	71	10	
98	2099	Aravind	IT	7	59	11	
99	2100	Bhargavi	ECE	3	65	15	

	Internal_Marks	Marks	Result
95	28	46	Pass
96	16	72	Pass
97	21	96	Pass
98	21	58	Pass
99	27	85	Pass

DATA INFORMATION

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Student_ID            100 non-null   int64
1   Name                  100 non-null   object
2   Department             100 non-null   object
3   Semester              100 non-null   int64
4   Attendance             100 non-null   int64
5   Assignment_Marks       100 non-null   int64
6   Internal_Marks         100 non-null   int64
7   Marks                 100 non-null   int64
8   Result                100 non-null   object
dtypes: int64(6), object(3)
memory usage: 7.2+ KB
None

```

MISSING VALUES

```

Student_ID      0
Name            0
Department      0
Semester        0
Attendance       0
Assignment_Marks 0
Internal_Marks  0
Marks           0
Result          0
dtype: int64

```

STATISTICAL ANALYSIS

```

Mean Marks : 58.86
Median Marks : 58.0
Standard Deviation : 21.985003979985994
Variance : 483.34039999999993
Maximum Marks : 98
Minimum Marks : 18

```

Attendance Statistics

```

Mean Attendance : 76.91
Maximum Attendance : 98
Minimum Attendance : 55

```

Mode of Marks

```

0    74
Name: Marks, dtype: int64

```

Department Wise Average Marks

```

Department
AI      57.04
CSE     55.64
ECE     61.88
IT      60.88
Name: Marks, dtype: float64

```

Semester Wise Average Marks

```

Semester

```

```

3      56.794118
5      63.030303
7      56.818182
Name: Marks, dtype: float64

```

```

Total Students : 100
Passed Students : 80
Failed Students : 20
Pass Percentage : 80.00%
Fail Percentage : 20.00%

```

```

Top Performer
Student_ID      2027
Name            Mohan
Department      IT
Semester        7
Attendance      72
Assignment_Marks 9
Internal_Marks  16
Marks           98
Result          Pass
Name: 26, dtype: object

```

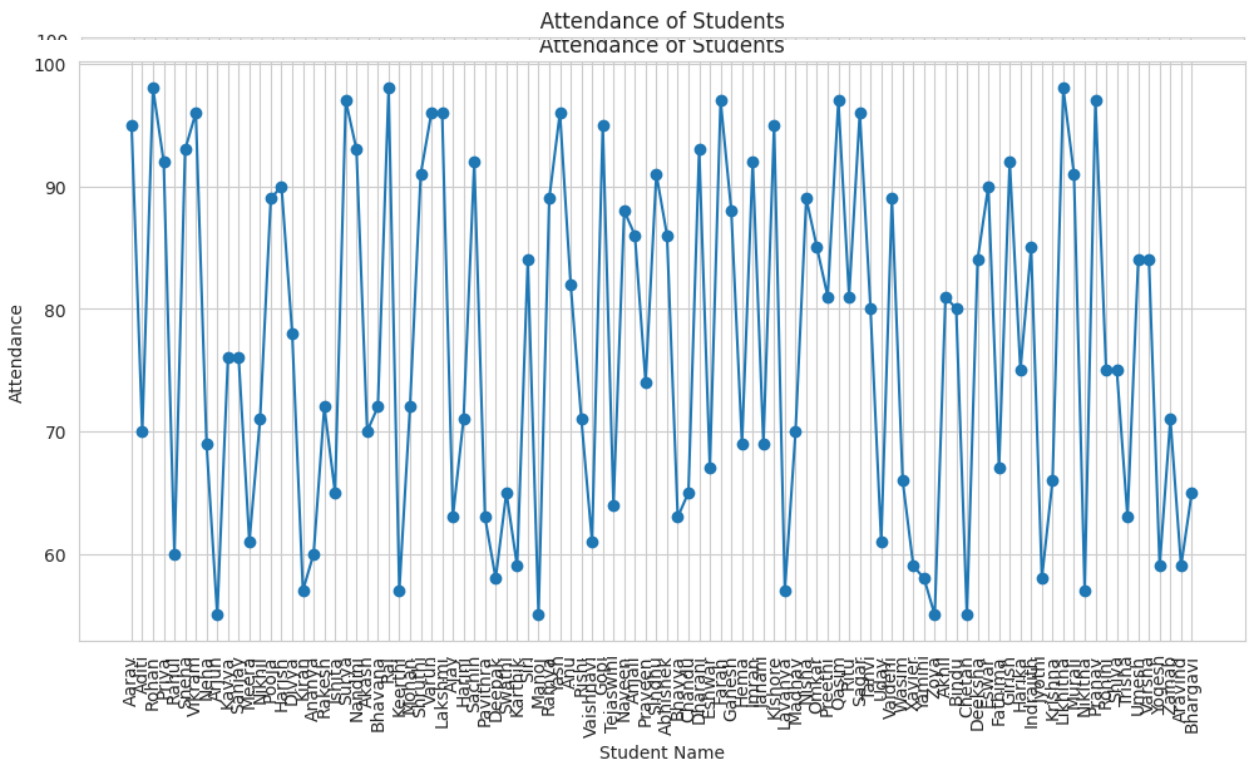
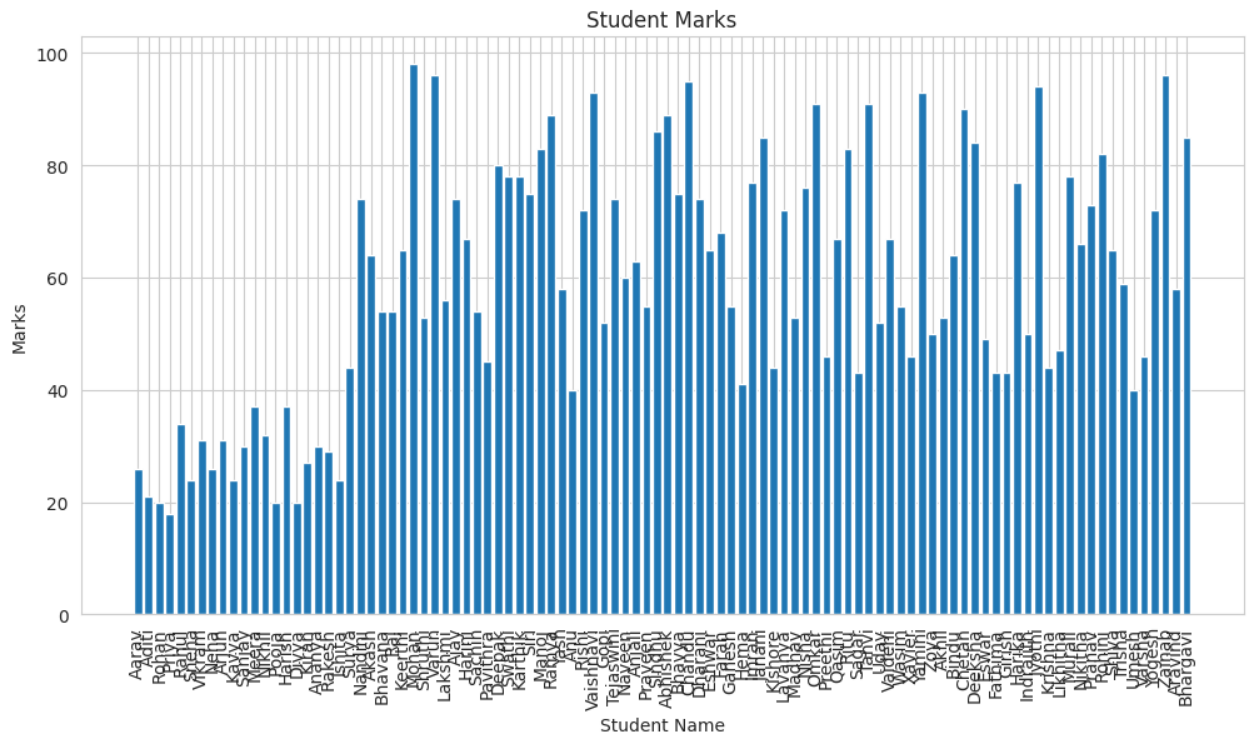
Students with Attendance Below 75%

	Student_ID	Name	Department	Semester	Attendance
Assignment_Marks \					
1	2002	Aditi	AI	5	70
11					
4	2005	Rahul	CSE	5	60
11					
7	2008	Neha	ECE	5	69
15					
8	2009	Arjun	CSE	7	55
20					
11	2012	Meera	ECE	7	61
13					
12	2013	Nikhil	CSE	3	71
20					
16	2017	Kiran	CSE	5	57
18					
17	2018	Ananya	AI	7	60
11					
18	2019	Rakesh	IT	3	72
15					
19	2020	Ishita	ECE	5	65
13					
22	2023	Akash	IT	5	70
10					
23	2024	Bhavana	ECE	7	72
18					
25	2026	Keerthi	AI	5	57
20					
26	2027	Mohan	IT	7	72
9					
30	2031	Ajay	IT	3	63
11					
31	2032	Harini	ECE	5	71
19					
33	2034	Pavithra	AI	3	63
16					

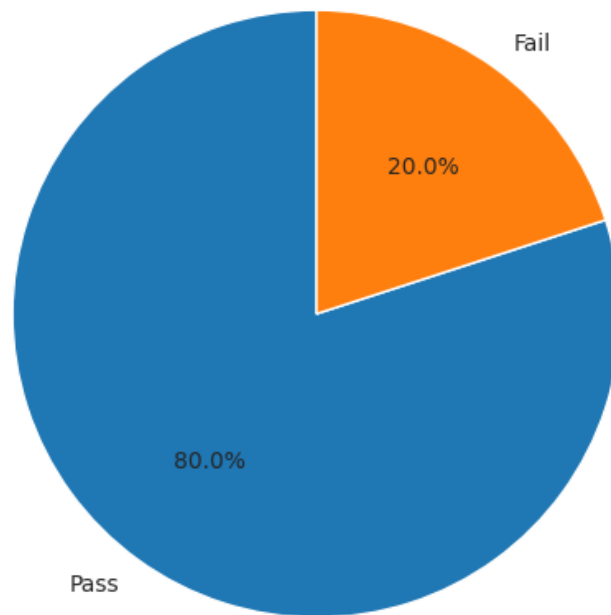
34	2035	Deepak	IT	5	58
9					
35	2036	Swathi	ECE	7	65
20					
36	2037	Karthik	CSE	3	59
14					
38	2039	Manoj	IT	7	55
18					
42	2043	Rishi	IT	3	71
16					
43	2044	Vaishnavi	ECE	5	61
18					
45	2046	Tejaswini	AI	3	64
13					
48	2049	Praveen	CSE	3	74
11					
51	2052	Bhavya	ECE	3	63
10					
52	2053	Chandu	CSE	5	65
12					
54	2055	Eshwar	IT	3	67
19					
57	2058	Hema	AI	3	69
9					
59	2060	Janani	ECE	7	69
8					
61	2062	Lavanya	AI	5	57
13					
62	2063	Madhav	IT	7	70
12					
70	2071	Uday	IT	5	61
11					
72	2073	Wasim	CSE	3	66
12					
73	2074	Xavier	AI	5	59
15					
74	2075	Yamini	IT	7	58
18					
75	2076	Zoya	ECE	3	55
9					
78	2079	Chetan	IT	3	55
14					
81	2082	Fathima	AI	3	67
12					
85	2086	Jyothi	AI	5	58
16					
86	2087	Krishna	IT	7	66
9					
89	2090	Nikitha	AI	7	57
17					
93	2094	Trisha	AI	3	63
18					
96	2097	Yogesh	CSE	3	59
16					
97	2098	Zainab	AI	5	71
10					
98	2099	Aravind	IT	7	59
11					
99	2100	Bhargavi	ECE	3	65
15					

	Internal_Marks	Marks	Result
1	14	21	Fail
4	17	34	Fail
7	28	26	Fail
8	15	31	Fail
11	21	37	Fail
12	11	32	Fail
16	17	27	Fail
17	13	30	Fail
18	30	29	Fail
19	21	24	Fail
22	24	64	Pass
23	27	54	Pass
25	20	65	Pass
26	16	98	Pass
30	27	74	Pass
31	28	67	Pass
33	25	45	Pass
34	14	80	Pass
35	23	78	Pass
36	22	78	Pass
38	13	83	Pass
42	15	72	Pass
43	19	93	Pass
45	15	74	Pass
48	11	55	Pass
51	25	75	Pass
52	26	95	Pass
54	19	65	Pass
57	20	41	Pass
59	12	85	Pass
61	12	72	Pass
62	25	53	Pass
70	16	52	Pass
72	24	55	Pass
73	27	46	Pass
74	27	93	Pass
75	17	50	Pass
78	18	90	Pass
81	16	43	Pass
85	12	94	Pass
86	29	44	Pass
89	12	66	Pass
93	30	59	Pass
96	16	72	Pass
97	21	96	Pass
98	21	58	Pass
99	27	85	Pass

Analysis Completed Successfully.



Pass and Fail Percentage



Marks Distribution

